NPC – UNIT 3 ASSIGNMENT

Configure a DNS Server and Web Server Using Cisco Packet Tracer

Submitted by	Vrushank Sabale
PRN	2125UCSM1092
Programme / Department	B.Tech CSE – Cybersecurity
Subject	NPC
Software	Cisco Packet Tracer 9.0.1

Practical Report

Submitted as part of the NPC Unit 3 practical assignment.

1. Aim and Objectives

Aim: To configure a DNS server and a web server in Cisco Packet Tracer, configure client PCs to use DNS, verify network connectivity, and access a website through its domain name.

Objectives

• Build a small LAN using two client PCs, a switch, a DNS server, and a web server.
• Assign unique IPv4 addresses to the devices.
• Enable DNS and create an A record for **www.college.com**.
• Enable HTTP on the web server.
• Test connectivity with ping and confirm domain-name resolution.
• Open the website in a client PC browser.

2. Devices and Network Design

The practical uses one Cisco 2960 switch to connect the two client PCs and both servers. All devices are on the same IPv4 subnet, so a router/default gateway is not required for the local tests in this assignment.

Device	Role	IPv4 Address	Subnet Mask
PC0	Client computer	192.168.1.10	255.255.255.0
PC1	Client computer	192.168.1.11	255.255.255.0
Server0	DNS server	192.168.1.2	255.255.255.0
Server1	Web server	192.168.1.3	255.255.255.0

DNS mapping: www.college.com → 192.168.1.3 (A record).
Client DNS setting: 192.168.1.2.

3. Topology and Device Setup

The devices were placed in the Packet Tracer logical workspace. The following screenshot documents the device layout before all links were completed.

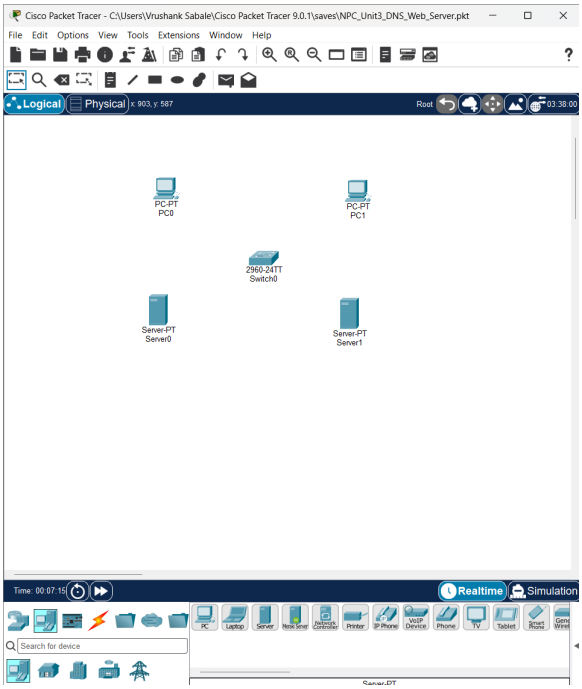


Figure 1. Device placement in the Packet Tracer workspace.

4. Network Connections and Server Configuration

The client PCs were connected to the Cisco 2960 switch using Copper Straight-Through Ethernet cables. The servers were also connected to the same switch for communication on the local subnet.

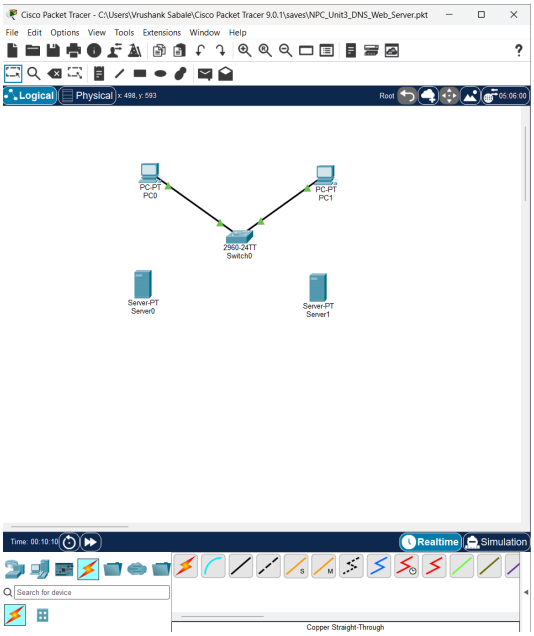


Figure 2. Client PCs connected to the switch during topology setup.

4.1 DNS Server IPv4 Configuration

Server0 was assigned the static IPv4 address 192.168.1.2 with subnet mask 255.255.255.0. No router is used for the local-network tests, so the default gateway is left at its default value.

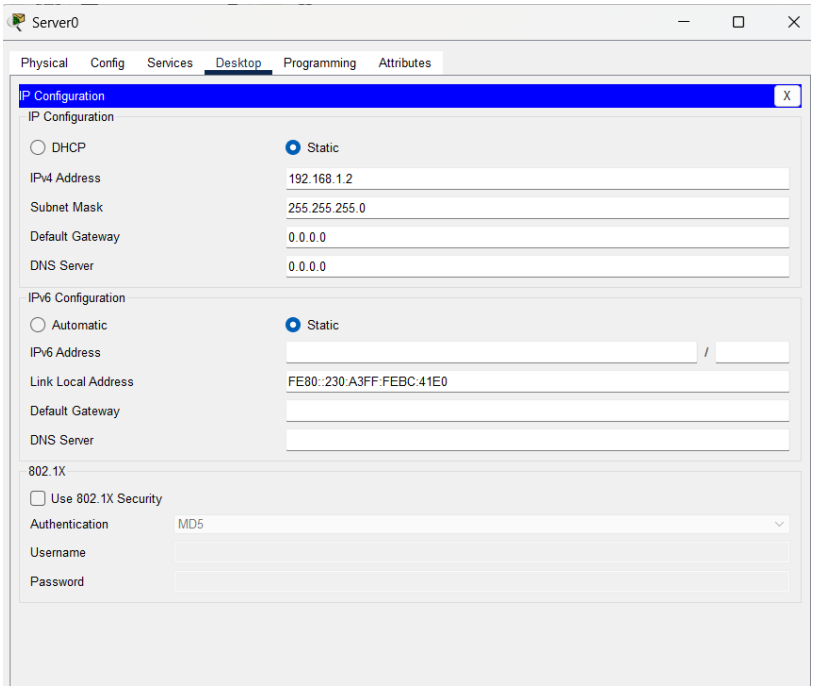


Figure 3. DNS server IPv4 configuration.

5. DNS Service Configuration

On Server0, the DNS service was switched On. An A record was added to map the domain name www.college.com to the web server address 192.168.1.3. This record allows clients configured to use 192.168.1.2 as their DNS server to resolve the domain name.

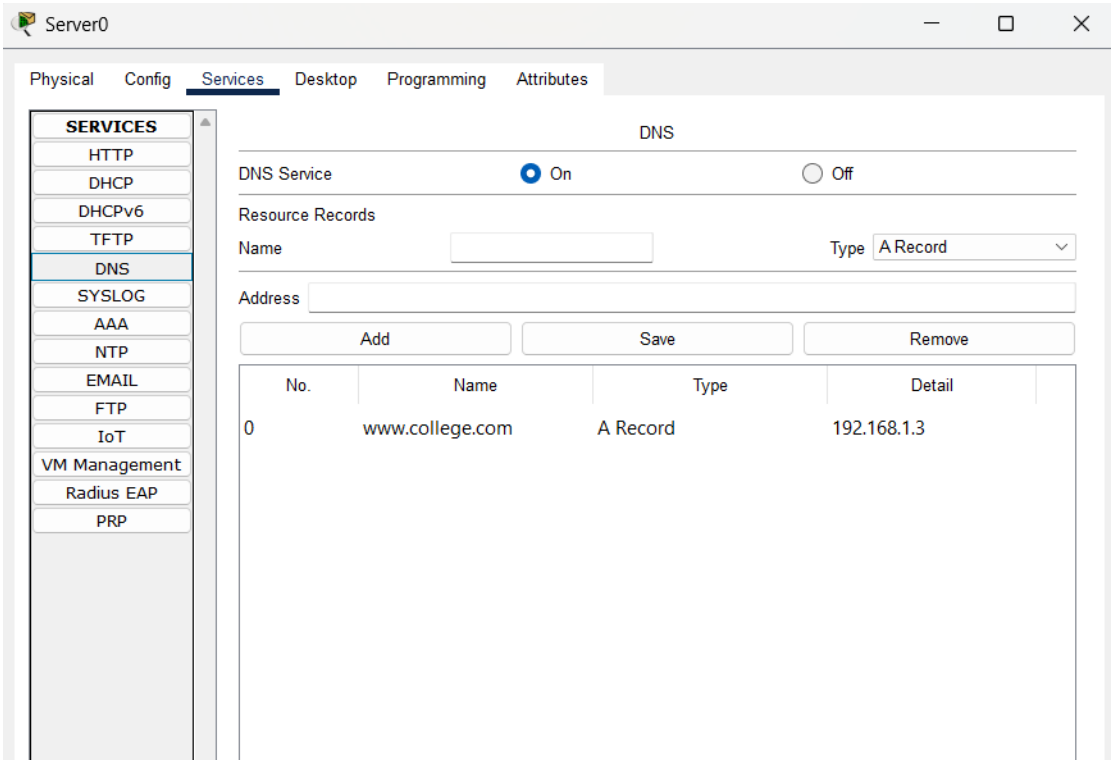


Figure 4. DNS service enabled and A record added.

6. Web Server Configuration

Server1 was assigned the static IPv4 address 192.168.1.3 with subnet mask 255.255.255.0. The HTTP service was enabled. HTTPS was also shown as enabled in the captured configuration screen, although HTTP is the service used for the required browser test.

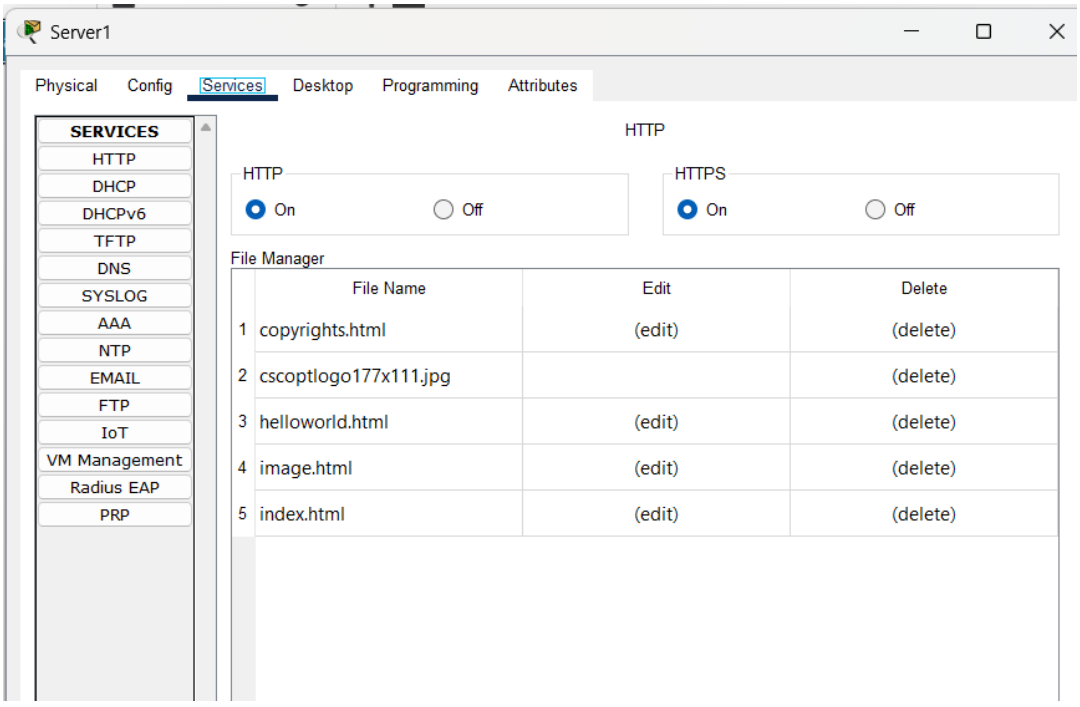


Figure 5. Web server HTTP service enabled.

7. Client Configuration and Testing

Both PCs were configured with unique addresses in the 192.168.1.0/24 network and the DNS server address 192.168.1.2. The final verification was performed from PC0 using the Packet Tracer command prompt.

7.1 Ping Tests and DNS Resolution

The successful test output shows replies from 192.168.1.2 (DNS server). The command ping www.college.com resolves the domain to 192.168.1.3 and receives replies from the web server. The displayed statistics report 4 packets sent, 4 received, and 0% packet loss for both tests.

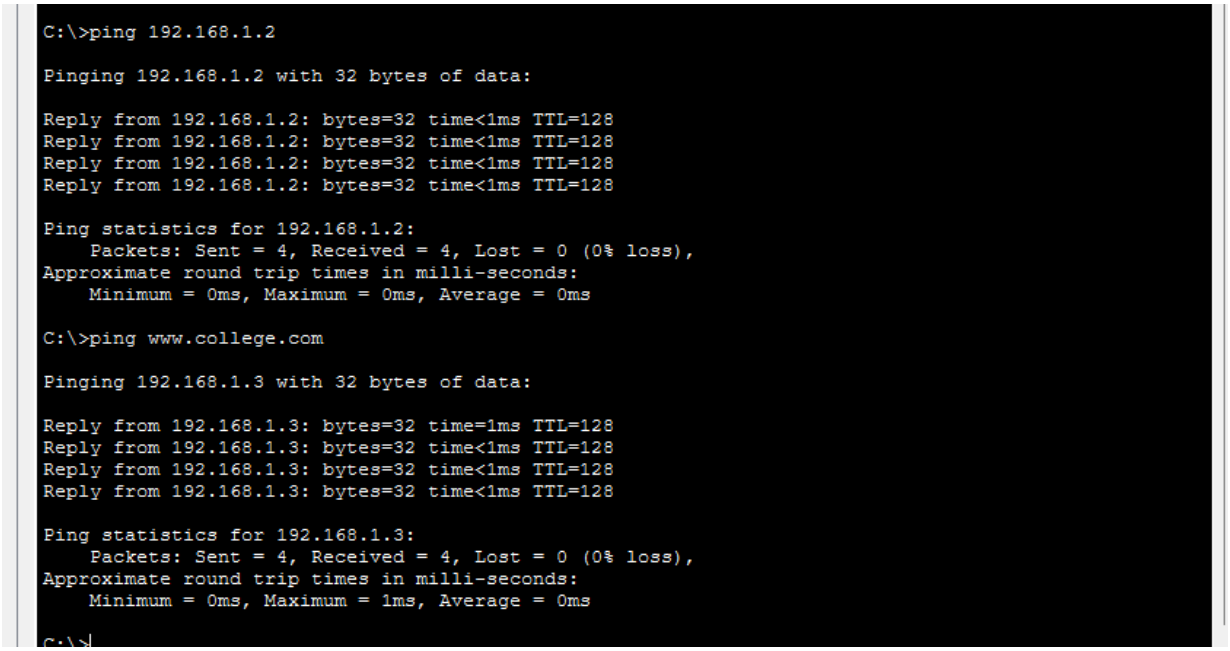


Figure 6. Successful connectivity and domain-name resolution tests from PC0.

8. Website Access Verification

The client browser was used to open the domain name rather than entering the server IP address directly. The default Packet Tracer homepage loaded successfully at http://www.college.com, confirming that DNS resolution and HTTP access worked together.

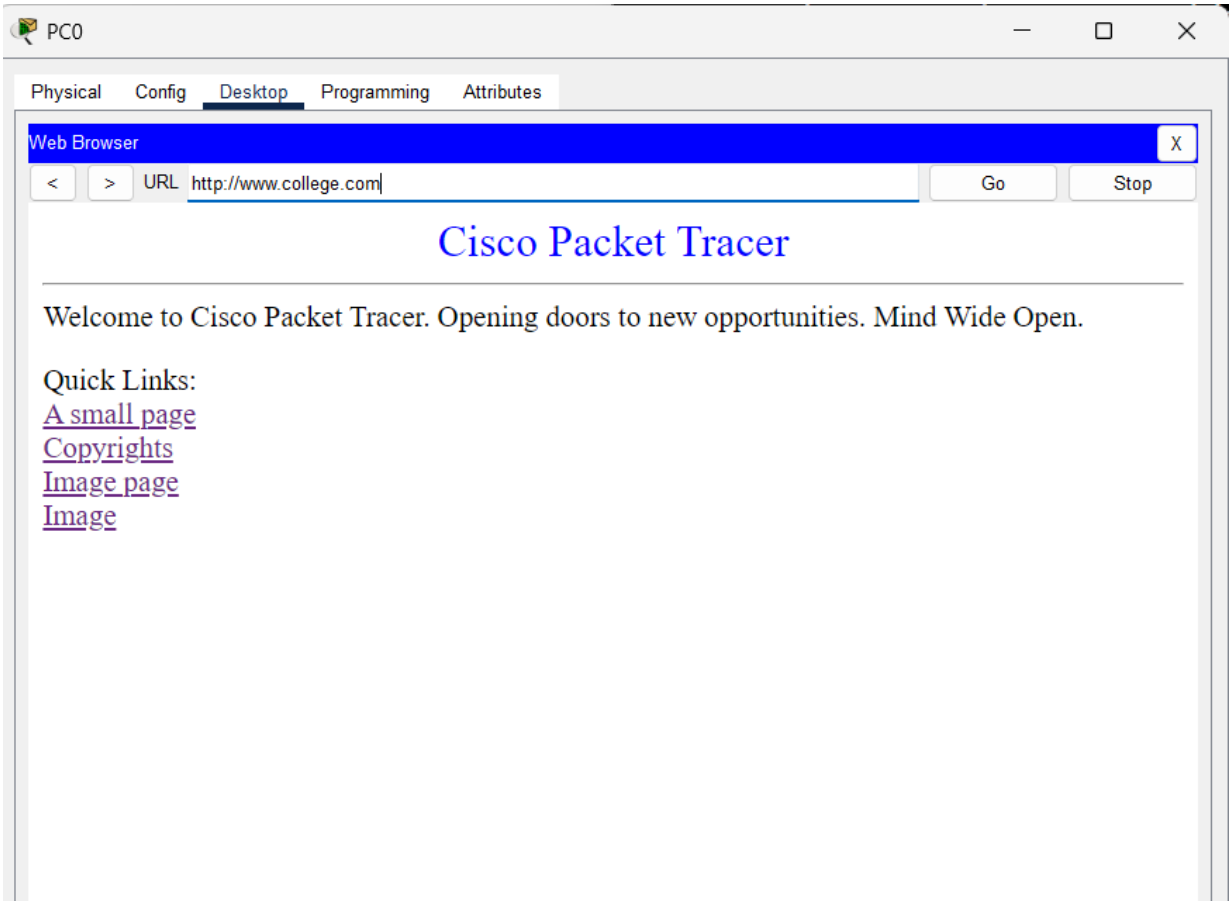


Figure 7. Homepage loaded using http://www.college.com.

An additional browser screenshot shows that a page hosted by the same web server was also reachable.

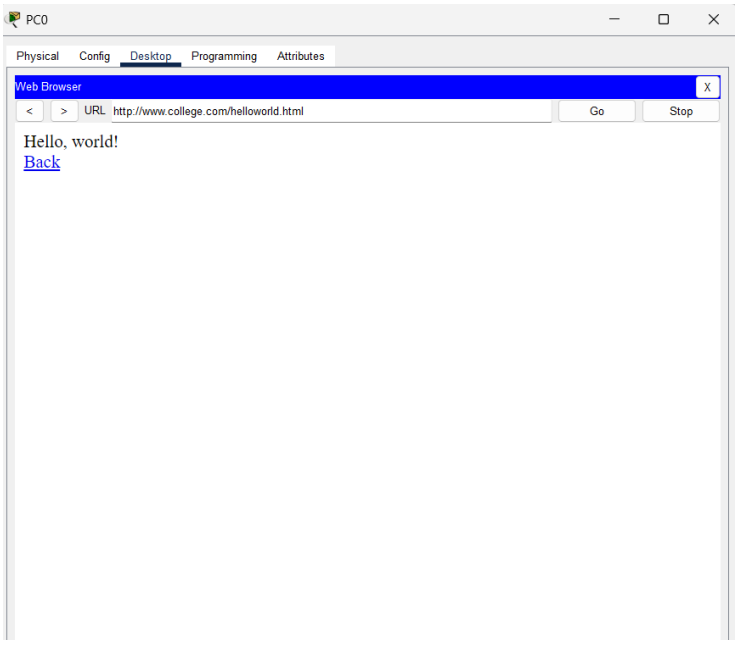


Figure 8. Web server page displayed in the browser.

9. Results

The practical requirements were demonstrated as follows:

- The DNS server at 192.168.1.2 was reachable from PC0.
- The web server at 192.168.1.3 was reachable from PC0.
- The domain www.college.com resolved to 192.168.1.3.
- The web server returned a webpage through the browser using the domain name.

10. Conclusion

The DNS and web server configuration was completed successfully in Cisco Packet Tracer. The DNS A record associated www.college.com with the web server's IPv4 address, and the client was able to resolve the domain and access the hosted webpage over HTTP. The ping tests verified local network connectivity between the client and the servers.

11. Key Configuration Summary

Item	Configured value
DNS server	192.168.1.2 / 255.255.255.0
Web server	192.168.1.3 / 255.255.255.0
DNS service	On
DNS A record	www.college.com → 192.168.1.3
HTTP service	On
Client DNS server	192.168.1.2
Verification	Ping successful; domain resolves; webpage loads

Submitted by: Vrushank Sabale | PRN: 2125UCSM1092